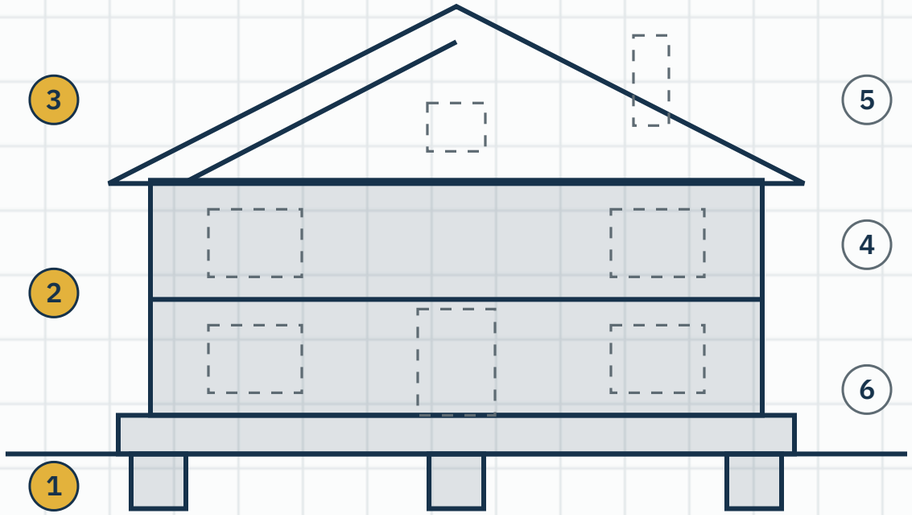
Veridom Milestones

Construction-loan draws, proven without handing over the paperwork.

A Veridom module on Midnight
Midnight Buildathon, wave 3



Every tranche costs the builder their privacy.

What the bank asks for at each draw

Site photos	copied
Invoices and amounts paid	copied
Contractors and their contracts	copied
Personal and family data	copied

The copies stay in the bank's systems and travel to surveyors. The builder keeps no say over them.

What the bank needs to decide

Is **stage *k* finished**?

Did an **inspector the bank trusts** confirm it?

Does the payout **stay within the approved loan**?

Three yes-or-no facts. None of them needs the documents themselves.

Facts on a shared ledger, files on the builder's device.

A shared database

Everyone has to trust whoever runs it, and it still holds every document.

A public blockchain

Verifiable, but every amount and every payment link would be readable by anyone, for ever.

Midnight

The public ledger holds outcomes and commitments. Private state holds the data. Zero-knowledge proofs connect the two, and selective disclosure opens one fact when someone needs it.

One stage, from site visit to payout.

1

The bank sets up the loan

Registers its inspectors, and the loan as a commitment to the limit and stage caps.

2

The builder accepts

Proves it holds exactly the terms the bank committed to.

3

The inspector signs

After the site visit: one signature over this contract, project, stage and evidence bundle.

4

The builder proves the stage

Stage *k* complete, signed by an authorised inspector, signature never used before.

5

The builder requests the tranche

Within the stage cap, and the running total within the loan limit. No amount is published.

6

The bank pays, the auditor checks

The bank checks the amount against its commitment. An auditor gets one fact, with consent on chain.

● Transaction with a ZK proof ○ Off chain

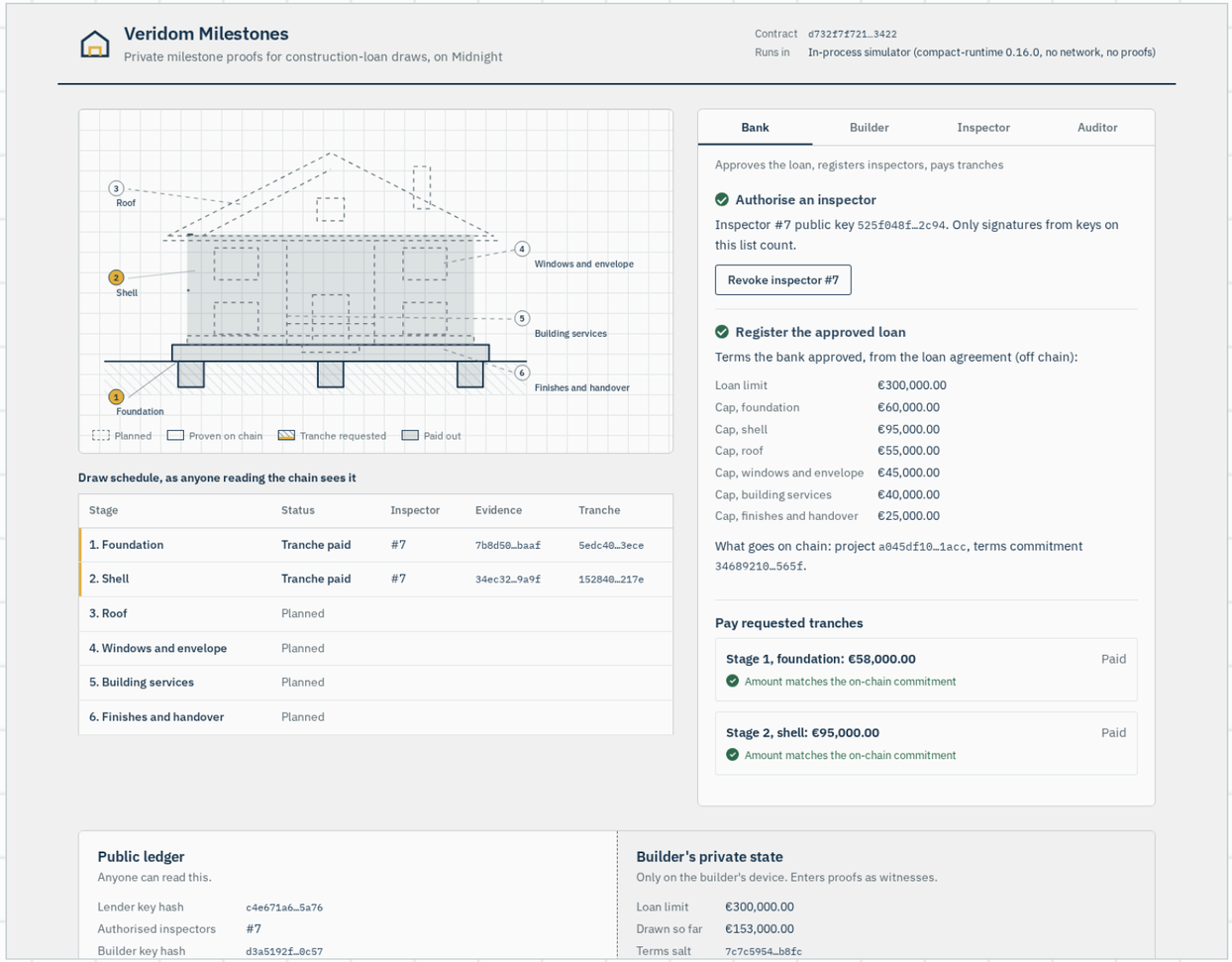
What the chain sees, and what it never sees.

Public ledger	Private state (witnesses)
Lender and builder key hashes	Lender and builder secrets
Terms commitment	Loan limit, stage caps, salt
Drawn-total commitment	Running total drawn
Stage *k*: status, inspector id	Evidence bundle hash, signature
Tranche commitment	Tranche amount
Spent-signature nullifiers	
Salted consent receipts	Who received which fact

Compact 0.31.1, 8 provable circuits. Inspector signatures are verified inside the proof, caller identity comes from witness secrets, and write-back witnesses keep the private running total.

```
// requestTranche, inside the proof
const terms = openTerms(pid, p);
const drawn = openDrawn(pid, p);
const amount = trancheAmount(pid, st);
assert(amount > 0, "...");
assert(amount <= capForStage(
        terms.stageCaps, st), "...");
const newTotal = drawn + amount;
assert(newTotal <= terms.limit,
  "Cumulative draw would exceed
   the approved loan limit");
```

It runs, with real proofs.



50

simulator tests pass: over-limit, rogue inspector, replay, no consent, leak scan

9

transactions with ZK proofs on a local Midnight devnet, about 25 s each

4

roles in one browser app: bank, builder, inspector, auditor; Lace-ready

Where Veridom stands today.

Veridom is live at **veridom.space**: it tracks a private home build against the approved plans, with a diary, photos and inspections.

Construction-industry specialists are testing it and have given positive feedback.

We are in talks with banks and investment firms.

The first pilot is the founder's own first build.

No paying users yet.

The lender pays; the builder keeps the data.

Who uses it

Banks and building societies that finance private home construction and pay it out stage by stage. Builders, through the Veridom app. Inspectors, with a signing tool.

Who pays

The lender, per active construction loan: fewer documents to collect, store and protect, and a verifiable record of every release. Builders use it free as part of Veridom.

How it spreads

Start with the lenders Veridom already talks to, on private builds. The same proof fits other staged payouts: completion checks for insurers, stage-based grants and subsidies.

From devnet to a first bank pilot.

Wave 3, done

- Compact contract, 8 circuits
- 50 simulator tests, devnet run with real proofs
- Four-role web app, Lace wiring

Next

- Deploy on Preprod through Lace
- Evidence bundles straight from the Veridom diary
- Inspector signing app with hardware-held keys

Then

- Pilot on the founder's own build
- Pilot with a lender from current talks
- Standard-library Schnorr once ledger 9 is live

The bank sees the facts. The builder keeps the files.

We are looking for a lender to pilot with, and for feedback from the Midnight team.

veridom.space
Open source under Apache-2.0